

IASI, Romania

WMO: 150900

Lat: 47.1636N

Lon: 27.6273E

Elev: 248

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.9	9.7	-0.6	5.4	4.9	5.6	7.4	11.4	24.9	23.4	20.9	26.9	4.7	290	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	92.2	69.7	89.1	68.8	86.2	67.9	73.3	85.3	71.4	83.6	69.8	81.6	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.6	110.0	78.2	67.8	103.1	75.8	66.2	97.3	74.1	37.0	85.3	35.3	83.6	34.0	81.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.0	13.7	-2.9	96.7	6.8	3.5	-7.8	99.2	-11.8	101.3	-15.6	103.3	-20.5	105.9		
			-3.3	76.2	6.8	2.3	-8.2	77.9	-12.2	79.3	-16.0	80.6	-20.9	82.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	27.8	31.8	40.9	52.6	62.7	69.2	72.6	71.6	62.3	51.4	41.6	31.0	(6)
(7)		DBStd	17.59	9.43	10.31	8.86	7.05	6.61	5.60	5.32	5.69	6.79	7.71	8.77	9.42	(7)
(8)		HDD50	2496	688	510	299	52	3	0	0	0	2	77	277	588	(8)
(9)		HDD65	5671	1153	929	747	373	122	23	6	10	128	425	702	1053	(9)
(10)		CDD50	3009	0	0	17	131	395	577	702	671	371	119	25	1	(10)
(11)		CDD65	712	0	0	0	3	50	150	243	216	47	3	0	0	(11)
(12)		CDH74	7149	0	0	0	65	576	1477	2361	2144	486	40	0	0	(12)
(13)		CDH80	2669	0	0	0	13	146	504	970	894	137	5	0	0	(13)
(14)	Wind	WSAvg	5.7	6.0	6.3	6.9	6.7	5.7	5.3	4.8	4.5	5.0	5.2	6.1	5.8	(14)
(15)	Precipitation	PrecAvg	22.8	1.2	1.1	1.3	1.9	2.4	3.5	3.0	2.1	1.9	1.4	1.4	1.3	(15)
(16)		PrecMax	32.3	6.3	2.6	3.6	4.3	8.1	11.5	8.0	6.4	7.2	6.0	4.4	3.3	(16)
(17)		PrecMin	14.6	0.1	0.1	0.1	0.0	0.1	0.5	0.5	0.0	0.0	0.1	0.0	0.0	(17)
(18)		PrecStd	4.2	0.9	0.7	0.9	1.1	1.6	2.0	1.8	1.5	1.9	1.2	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	59.6	69.6	81.3	88.8	91.8	97.4	97.3	88.7	78.8	68.7	55.8	(19)
(20)			MCWB	43.5	49.3	52.5	61.6	65.8	70.2	68.8	70.4	65.8	63.1	56.8	49.2	(20)
(21)		2%	DB	45.5	52.5	64.1	74.7	83.9	88.5	92.8	92.2	84.0	73.2	62.1	50.1	(21)
(22)			MCWB	41.0	45.2	50.7	58.0	64.2	68.8	69.7	69.8	64.1	60.1	54.5	46.2	(22)
(23)		5%	DB	42.0	47.9	59.1	70.4	80.4	85.7	89.0	88.8	79.7	68.2	57.5	45.8	(23)
(24)			MCWB	39.1	42.7	48.2	55.7	63.2	68.1	69.6	68.7	62.9	57.9	52.1	42.9	(24)
(25)		10%	DB	39.3	44.2	54.6	66.4	76.7	82.6	85.6	85.3	75.4	63.9	53.8	42.2	(25)
(26)			MCWB	37.3	40.4	45.6	53.6	61.7	66.5	68.5	67.8	61.8	56.1	50.1	39.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	49.9	54.5	62.2	69.2	74.5	75.9	75.6	69.1	64.8	58.4	50.4	(27)
(28)			MCDWB	49.5	57.4	66.2	78.1	81.7	85.9	87.4	88.5	83.2	75.5	66.1	53.6	(28)
(29)		2%	WB	41.7	46.0	51.5	59.4	66.7	71.9	73.4	72.9	66.5	61.6	55.5	46.5	(29)
(30)			MCDWB	44.6	50.7	62.1	71.7	79.3	83.6	85.0	85.7	78.1	69.6	60.6	49.9	(30)
(31)		5%	WB	39.4	43.4	49.1	57.2	64.7	69.9	71.6	70.8	64.6	59.1	53.2	43.1	(31)
(32)			MCDWB	41.5	47.0	57.9	67.5	76.7	81.2	83.4	83.5	75.2	66.6	57.1	45.5	(32)
(33)		10%	WB	37.2	40.6	46.6	55.0	62.9	68.2	70.0	69.0	62.9	56.7	50.5	40.0	(33)
(34)			MCDWB	39.2	43.9	53.1	63.9	73.6	79.0	81.3	80.9	72.9	62.7	53.6	41.9	(34)
(35)	Mean Daily Temperature Range		MDBR	10.3	12.5	16.9	20.1	21.6	21.0	21.4	22.3	20.3	17.5	12.3	10.4	(35)
(36)		5% DB	MCDBR	13.2	18.6	25.1	27.7	27.2	25.6	26.9	28.1	27.3	25.1	18.5	15.2	(36)
(37)			MCWBR	9.7	12.3	14.1	13.4	11.9	10.4	9.5	9.8	10.8	13.2	11.3	11.4	(37)
(38)		5% WB	MCDBR	12.0	17.3	23.3	24.2	24.4	22.8	22.8	24.5	23.5	22.4	16.3	14.3	(38)
(39)			MCWBR	9.4	12.1	13.6	12.4	11.9	10.5	10.3	10.1	10.7	12.4	10.6	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.357	0.392	0.449	0.421	0.418	0.446	0.453	0.410	0.376	0.352	0.323	(40)	
(41)		taud	2.265	2.230	2.209	2.083	2.240	2.281	2.209	2.180	2.259	2.333	2.335	2.313	(41)	
(42)		Ebn at Noon	233	251	260	255	268	269	259	251	251	242	223	220	(42)	
(43)		Edh at Noon	25	32	38	48	43	41	43	43	36	29	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	379	619	1006	1407	1799	1890	1891	1712	1204	769	392	295	(44)	
(45)		RadStd	55	87	103	123	150	162	143	125	137	92	37	39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.26	N/A	N/A	N/A	N/A	N/A	-239	-349	+184	+92	
(47)	Regional (33 neighbors)	+1.21	N/A	N/A	+1.37	N/A	N/A	-259	-360	+182	+67	(47)

Nomenclature: See separate page